CSI4901 Capstone Project

ADAPTIVE SELF-HEALING FRAMEWORK FOR QUANTUM MACHINE LEARNING

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ABSTRACT

Variational Quantum Circuits (VQCs) serve as the primary computational backbone for near-term Quantum Machine Learning (QML) in the Noisy Intermediate-Scale Quantum (NISQ) era. However, hybrid quantum-classical training loops are severely constrained by silent, non-crashing training failuresâ€”including exponentially vanishing gradients (barren plateaus), hardware and decoherence noise-induced landscape flattening, optimizer instability, excessive circuit depth, and suboptimal feature-map encodings or parameter initializations. Existing mitigation approaches address these bottlenecks strictly in isolation through static, design-time interventions rather than dynamic runtime recovery. To overcome these critical barriers, this project proposes an **Adaptive Self-Healing Framework for Quantum Machine Learning** centered on a novel, patent-oriented **Adaptive Repair Selection Algorithm (ARSA)**. The proposed system introduces a closed-loop, 15-module autonomic architecture that continuously monitors VQC training telemetry, detects abnormal training degradation, and computes a calibrated posterior distribution over candidate root causes. An entropy-driven **Diagnosis Confidence Gate** governs diagnostic escalation, triggering supplementary diagnostic passes or falling back to lowest-risk historical repairs when diagnostic uncertainty is high. The framework encodes multi-dimensional failure telemetry into a **Failure Embedding Memory** for cosine-similarity retrieval of historical failure-repair outcomes, generates candidate repairs across six discrete repair arms (optimizer modification, circuit-depth reduction, feature-map modification, learning-rate adjustment, noise mitigation, and warm restart), and enforces a deterministic **Safety and Oscillation Constraint Filter** to suppress cyclic $(A \rightarrow B \rightarrow A)$ repair loops. Surviving candidates are evaluated pre-deployment within a **Noise-Aware Digital Twin Simulation Proxy** and scored via a scalarized **Multi-Objective Performance Predictor** balancing accuracy gain, convergence speed, circuit depth/shot overhead, and stability. Crucially, **Contextual Thompson Sampling (CTS)** adaptively selects the optimal repair arm under uncertainty, deploying the repair with automatic checkpointing and degradation-triggered rollback while continually updating Bayesian posteriors and the historical knowledge base.

Keywords: *Quantum Machine Learning, Variational Quantum Circuits, Autonomous Self-Healing Systems, Contextual Thompson Sampling, Adaptive Repair Selection Algorithm (ARSA), Barren Plateaus, Probabilistic Fault Diagnosis, Noise-Aware Digital Twin.*

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Quantum Machine Learning (QML) has emerged as one of the most promising computational paradigms for Noisy Intermediate-Scale Quantum (NISQ) processors [1, 2, 10]. At the core of near-term QML are Variational Quantum Circuits (VQCs)â€”also referred to as Parameterized Quantum Circuits (PQCs) or Quantum Neural Networks (QNNs)â€”which operate within a hybrid quantum-classical optimization loop [1, 3, 16]. In a typical VQC pipeline, classical input features are embedded into a high-dimensional quantum Hilbert space via a unitary feature map [4, 5, 17], processed through layers of parameterized single-qubit rotations and multi-qubit entangling gates, and measured via Pauli observables to evaluate an empirical loss function. A classical optimizer (such as Adam, SPSA, or Quantum Natural Gradient [12, 13]) then iteratively updates the circuit parameters using gradients estimated via the parameter-shift rule [6, 14].

Despite strong theoretical expressibility and potential representational advantages [8, 31, 32], training VQC models in practice is notoriously fragile. Unlike traditional software defects that raise explicit runtime exceptions or syntax errors [51], VQC training failures occur *silently* within the optimization dynamics. A VQC training run may stall, diverge, or collapse due to an intricate interaction of algorithmic, architectural, and physical factors: (i) exponentially vanishing gradients caused by random initialization or deep circuit expressibility (barren plateaus) [18, 23, 33]; (ii) cost-function-induced landscape flatness [19]; (iii) hardware decoherence and gate noise that exponentially flatten the loss surface (noise-induced barren plateaus) [25]; (iv) spurious local minima traps [24]; (v) ill-conditioned learning rates or optimizer mismatch [12, 13]; and (vi) poor feature-map frequency encoding [7, 17].

Currently, when VQC training degrades, recovery follows a slow, manual paradigm: *Detect → Human Investigates → Human Fixes*. Practitioners manually inspect loss curves, guess hyperparameter or ansatz modifications, and restart training from scratch. Existing research addresses individual symptoms in isolationâ€”such as structured initialization [20], layerwise training [21], design-time architecture search [34, 37], or static quantum error mitigation [26, 28, 29]â€”without an integrated, closed-loop self-healing mechanism capable of autonomously monitoring, diagnosing, simulating, and repairing failing QML models during execution.

1.2 MOTIVATION

This capstone project (Project ID: **20127IPG02**) is motivated by four fundamental engineering and intellectual-property drivers aimed at developing a novel, potentially patentable system and method for autonomous QML recovery:

1. **Silent Multi-Factorial Degradation in VQC Training:** In QML, identical observable symptoms (such as a plateaued training loss) can stem from fundamentally different underlying root

causesâ€”for example, an overly deep entangling ansatz [18, 33], an excessive learning rate causing parameter oscillation, or depolarizing channel noise [25]. Applying a static heuristic fix (such as reducing learning rate when the circuit is trapped in a barren plateau) wastes scarce quantum shot budgets and fails to restore trainability.

2. **Need for Confidence-Gated Probabilistic Diagnosis and Pre-Deployment Verification:** Because quantum gradient measurements are inherently stochastic due to finite shot noise and hardware fluctuations [11, 13], single-pass fault diagnosis carries significant uncertainty. Acting blindly on low-confidence diagnoses can corrupt partially trained circuit weights. A self-healing system must explicitly quantify diagnostic confidence (via normalized posterior entropy), trigger supplementary diagnostic passes when uncertainty is high, and screen candidate repairs in a noise-aware simulation proxy (Digital Twin) [56, 57] before touching the live QML model.

3. **Preventing Oscillatory Repair Loops and Catastrophic Regression:** Naive automated repair loops readily fall into cyclic oscillation (e.g., applying Repair $A_1 \rightarrow$ Repair $A_2 \rightarrow$ Repair A_1) or deploy destructive modifications that degrade classification accuracy. Integrating an explicit Safety and Oscillation Constraint Filter alongside pre-deployment state checkpointing and automatic rollback is essential for operational reliability.

4. **Contextual Sequential Decision-Making and Patent-Oriented Innovation:** Because no single repair action works universally across all circuit topologies, noise regimes, and training stages, repair selection must be formulated as a contextual sequential decision problem under uncertainty. By combining Failure Embedding Memory [53, 55], Digital-Twin multi-objective evaluation [58, 59], and **Contextual Thompson Sampling (CTS)** [60, 62] into the proposed **Adaptive Repair Selection Algorithm (ARSA)**, the system moves QML from manual trial-and-error to autonomous closed-loop self-healing: *Monitor → Diagnose → Select Repair → Verify → Learn*.

1.3 SCOPE OF THE PROJECT

The technical scope of the proposed Adaptive Self-Healing Framework for Quantum Machine Learning encompasses the following architectural boundaries, operational conditions, and development parameters:

- **Locked 15-Module Closed-Loop Architecture:** The framework integrates (1) Quantum ML Model (VQC), (2) Continuous Health Monitoring, (3) Failure Detection, (4) Probabilistic Root-Cause Diagnosis, (5) Diagnosis Confidence Estimation & Confidence Gate, (6) Failure Embedding Memory Retrieval, (7) Candidate Repair Generation, (8) Safety & Oscillation Constraint Filter, (9) Digital Twin / Noise-Aware Simulation Proxy, (10) Multi-Objective Performance Prediction, (11) Adaptive Repair Selection Policy via Contextual Thompson Sampling (ARSA), (12) Safe Repair Deployment & Rollback, (13) Performance & Regret Evaluation, (14) Online Policy Update, and (15) Continual Learning Knowledge Base.
- **Candidate Repair Action Space (A_1â€“A_6):** The initial repair library comprises six discrete, parameterized repair arms: A_1 (Optimizer Modification, e.g., Adam $\leftrightarrow$

SPSA/QNG), A_2 (Circuit-Depth / Layer Reduction), A_3 (Feature-Map / Encoding Modification), A_4 (Learning-Rate Adjustment), A_5 (Noise Mitigation Activation/Tuning, e.g., ZNE), and A_6 (Structured Warm Restart / Parameter Reinitialization).

- **Thin-Vertical-Slice Prototyping & Datasets:** Initial implementation targets a reproducible PennyLane [9] VQC classifier trained on benchmark datasets (beginning with Iris and Wine/Breast Cancer dimensionality-reduced quantum feature encodings) under controlled failure injection (excessive learning rate, deep circuit barren plateaus, depolarizing noise channels, and bad initialization).
- **Patent-First Technical Orientation:** All literature synthesis, prior-art matrices, system architectures, and mathematical formulations are structured to support both VIT Review-1 academic requirements and subsequent provisional patent specification drafting (subject to formal patentability examination).

CHAPTER 2
PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

A comprehensive literature and prior-art investigation of **73 peer-reviewed journal and conference publications** (including over 50 high-impact journal articles across *Nature*, *Nature Communications*, *Nature Reviews Physics*, *Physical Review Letters*, *PRX Quantum*, *Quantum*, *Reviews of Modern Physics*, *ACM Computing Surveys*, and *IEEE Transactions*) alongside 7 patent documents was conducted. The literature is synthesized across six core thematic categories below, followed by the exhaustive 50-paper comparative prior-art table (Table 2.1) and patent prior-art mapping.

2.1.1 Category A: Quantum Machine Learning, VQCs & Expressibility [1]â€“[11], [16]â€“[17], [31]â€“[33], [66]â€“[73]

Foundational QML paradigms established by Biamonte et al. [2], Cerezo et al. [1], Peruzzo et al. [3], and Mitarai et al. [16] define hybrid quantum-classical learning over parameterized unitaries $U(\mathbf{x}, \boldsymbol{\theta})$. Havlicek et al. [4] and Schuld & Killoran [5] demonstrated how quantum feature maps project data into reproducing kernel Hilbert spaces, while Perez-Salinas et al. [7] and Schuld et al. [17] proved that data re-uploading and encoding topology directly control the Fourier frequency spectrum learnable by a VQC. Sim et al. [31], Nakaji & Yamamoto [32], and Hubregtsen et al. [70] introduced quantitative descriptors of ansatz expressibility (via Kullback-Leibler divergence from the Haar measure) and entangling capability (Meyer-Wallach measure). However, as Holmes et al. [23, 33] proved, excessive expressibility directly suppresses gradient varianceâ€”establishing that circuit architecture properties serve as vital diagnostic telemetry but require dynamic runtime adaptation when training stalls.

2.1.2 Category B: Barren Plateaus, Optimization Traps & Trainability [12]â€“[15], [18]â€“[24]

McClean et al. [18] proved that for deep random PQCs forming approximate 2-designs, the variance of the cost gradient $\text{Var}[\partial_k C]$ vanishes exponentially as $\mathcal{O}(2^{-n})$. Cerezo et al. [19] extended this finding to prove that global cost observables induce barren plateaus even at shallow $\mathcal{O}(\log n)$ depths, whereas local observables preserve polynomial gradient scaling. Furthermore, Bittel & Kliesch [15] proved VQA optimization is NP-hard in general, and Anschuetz & Kiani [24] showed underparameterized VQCs suffer from ubiquitous spurious local minima traps. Existing remediesâ€”such as identity-block initialization by Grant et al. [20], layerwise learning by Skolik et al. [21], Quantum Natural Gradient by Stokes et al. [12], and stochastic parameter-shift rules [13, 14]â€”are strictly static design-time choices. None detect gradient collapse during training or dynamically switch optimizers/ansatze to recover.

2.1.3 Category C: Noise-Induced Degradation & Quantum Error Mitigation [25]â€“[30]

Physical NISQ execution introduces decoherence, depolarizing gate errors, and readout bias [10, 11]. Wang et al. [25] proved that local Pauli noise channels cause noise-induced barren plateaus (NIBPs),

flattening the entire cost landscape exponentially with circuit depth L regardless of initialization or cost locality. To counteract hardware noise without full fault-tolerant error correction, Temme et al. [26], Li & Benjamin [27], and Endo et al. [28] formulated Zero-Noise Extrapolation (ZNE) and Probabilistic Error Cancellation (PEC), validated experimentally by Kandala et al. [29] and Kim et al. [30]. However, continuous error mitigation multiplies quantum circuit sampling overhead by factors of $3\times$ to $10\times$. Current systems apply QEM statically rather than selectively activating or tuning mitigation (A_5) only when noise-induced degradation is diagnosed.

2.1.4 Category D: Quantum Circuit Architecture Search & Program Repair [34]â€“[40], [51]â€“[52]

Dynamic circuit modification has been explored via ADAPT-VQE by Grimsley et al. [35], Quantum Circuit Architecture Search (QCAS) by Du et al. [34], RL-based VQC pruning by Ostaszewski et al. [36], and noise-adaptive co-search (QuantumNAS) by Wang et al. [37]. In quantum software engineering, Paltenghi & Pradel [51] cataloged empirical platform bugs, and Usandizaga et al. [52] recently introduced gate-prioritization repair for faulty unitary programs. Yet, architecture search methods [34, 37] operate as expensive outer-loop searches from scratch, while quantum program repair [52] fixes discrete gate bugs against unit tests rather than healing continuous VQC optimization failures.

2.1.5 Category E: Autonomic Self-Healing, Fault Diagnosis & Anomaly Detection [41]â€“[50]

Self-healing computing originates from Kephart & Chess's Autonomic Computing architecture [41] and the Monitor-Analyze-Plan-Execute-Knowledge (MAPE-K) control loop [42, 43, 44]. Time-series anomaly detection [47, 49], concept-drift adaptation [50], and ML-based root-cause analysis [48] provide classical foundations for health telemetry. Most relevantly, in classical deep learning, Zhang et al. [45] developed *AutoTrainer* and Wardat et al. [46] developed *DeepDiagnosis* to automatically detect vanishing/exploding gradients or oscillating loss in classical DNNs and apply rule-based fixes. However, classical DNN self-healing systems lack awareness of quantum Hilbert-space expressibility, shot noise, barren plateaus, or quantum noise channels, and rely on rigid deterministic rules without confidence gating or pre-deployment simulation.

2.1.6 Category F: Failure Memory, Digital Twins, Multi-Objective & Contextual Bandits [53]â€“[65]

To avoid memoryless trial-and-error, Case-Based Reasoning (CBR) [53], Experience Replay [54], continual learning [65], and vector similarity indexing (FAISS) [55] enable systems to retrieve historical experiences. Digital Twins [56, 57] provide virtual simulation proxies for risk-free "what-if" screening prior to live deployment, while multi-objective decision frameworks [58, 59] formalize trade-offs across competing performance criteria. Crucially, for sequential action selection under uncertainty, Contextual Bandits [61, 64] and **Contextual Thompson Sampling (CTS)**â€”formally analyzed by Agrawal & Goyal [62], Russo et al. [60], and Chapelle & Li [63]â€”achieve provable $\mathcal{O}(d\sqrt{T\ln T})$ Bayesian regret bounds by sampling from arm posteriors conditioned on context $\mathbf{x}_t$. Synthesizing these mechanisms into a unified QML self-healing loop forms the core inventive premise of our project.

2.1.7 Extended Comparative Analysis Table of Surveyed Literature

Table 2.1 presents a structured comparative prior-art synthesis of 50 core research papers from our 73-paper survey, detailing each work's methodology and specific technical limitation relative to our proposed Adaptive Self-Healing Framework.

Table 2.1: Comprehensive Analysis of 50 Key Research Papers Across Thematic Categories

Ref	Authors	Year	Venue	Key Contribution	Identified Limitation / Gap Relative to Proposed Study
[45]	Zhang et al.	2021	IEEE/ACM ICSE	AutoTrainer: Automated DNN training problem detection and repair via monitoring + dynamic healing.	Classical DNNs only; fixed rule-based repair actions without contextual bandits.
[46]	Wardat et al.	2022	IEEE/ACM ICSE	DeepDiagnosis: Automated fault localization and actionable fix recommendation in deep learning.	Restricted to classical deep networks; lacks quantum noise/landscape awareness.
[18]	McClean et al.	2018	Nature Commun.	Proves gradients in random parameterized quantum circuits vanish exponentially with qubit count.	Theoretical characterization only; provides no runtime detection or autonomous repair.
[25]	Wang et al.	2021	Nature Commun.	Proves hardware Pauli/depolarizing noise induces barren plateaus exponentially in circuit depth.	Analyzes noise degradation without an active runtime recovery loop.
[19]	Cerezo et al.	2021	Nature Commun.	Shows global cost functions induce barren plateaus even in shallow parameterized quantum circuits.	Preventive cost design only; no automated runtime corrective switching.
[37]	Wang et al.	2022	IEEE HPCA	QuantumNAS: Noise-adaptive co-search for robust variational quantum circuits and qubit mappings.	Heavy design-time search rather than lightweight runtime failure recovery.
[34]	Du et al.	2020	arXiv / PRX	Quantum Circuit Architecture Search (QCAS) for error mitigation and trainability enhancement.	Lacks continuous health monitoring, confidence gating, or rollback protection.
[36]	Ostaszewski et al.	2021	NeurIPS	Reinforcement learning for dynamic optimization of variational quantum circuit architectures.	Optimizes architecture from scratch rather than healing mid-training failures.
[35]	Grimsley et al.	2019	Nature Commun.	ADAPT-VQE: Dynamically grows variational ansatze one operator at a time using gradient signals.	Domain-specific to molecular Hamiltonians; no multi-fault diagnostic taxonomy.
[24]	Anschuetz & Kiani	2022	Nature Commun.	Demonstrates underparameterized VQAs are swamped with spurious local minima traps.	Identifies landscape traps theoretically without automated escape/warm-restart policies.
[20]	Grant et al.	2019	Quantum	Identity-block parameter initialization strategy to avoid barren plateaus at start of training.	Static initialization prior to training; cannot recover once optimization stalls mid-run.
[21]	Skolik et al.	2021	Quantum Mach. Intell.	Layerwise learning strategy that incrementally trains shallow circuit blocks to preserve gradients.	Fixed training schedule; lacks anomaly-triggered adaptation.

Ref	Authors	Year	Venue	Key Contribution	Identified Limitation / Gap Relative to Proposed Study
[31]	Sim et al.	2019	Adv. Quantum Technol.	Formalizes expressibility and entangling capability descriptors for parameterized quantum circuits.	Static benchmark metrics; no closed-loop failure recovery.
[32]	Nakaji & Yamamoto	2021	Quantum	Analyzes expressibility of alternating layered ansatze and its connection to gradient variance.	Analytical study only; lacks runtime monitoring or repair selection.
[33]	Holmes et al.	2022	PRX Quantum	Establishes fundamental trade-off between ansatz expressibility and gradient magnitudes.	Does not provide an automated mechanism to dynamically constrain expressibility.
[12]	Stokes et al.	2020	Quantum	Introduces Quantum Natural Gradient (QNG) using the Fubini-Study metric tensor.	Selected statically beforehand; incurs high quantum metric estimation overhead.
[13]	Sweke et al.	2020	Quantum	Proves convergence properties of stochastic gradient descent with finite measurement shots in VQAs.	Does not adapt optimizer or learning rate when instability occurs.
[14]	Wierichs et al.	2022	Quantum	General parameter-shift rules for exact gradient evaluation on quantum processors.	Focuses purely on gradient calculation rather than gradient-health recovery.
[17]	Schuld et al.	2021	Phys. Rev. A	Proves data-encoding feature maps determine the Fourier frequency spectrum of VQC models.	Static encoding analysis; no runtime feature-map repair.
[15]	Bittel & Kliesch	2021	Phys. Rev. Lett.	Proves that classical optimization of variational quantum algorithms is NP-hard in general.	Complexity-theoretic proof without practical recovery heuristics.
[26]	Temme et al.	2017	Phys. Rev. Lett.	Introduces Zero-Noise Extrapolation (ZNE) and Probabilistic Error Cancellation (PEC).	Applied statically with high sampling overhead; not triggered selectively by diagnosis.
[28]	Endo et al.	2018	Phys. Rev. X	Develops practical exponential/polynomial extrapolation and error mitigation protocols.	Increases circuit execution overhead uniformly across all training steps.
[29]	Kandala et al.	2019	Nature	Experimental demonstration that error mitigation extends computational reach of noisy hardware.	Hardware-specific manual calibration; lacks autonomous policy selection.
[30]	Kim et al.	2023	Nature	Demonstrates utility-scale quantum computation on 127-qubit processor using ZNE/PEA.	Focuses on expectation estimation rather than iterative QML training stability.
[51]	Paltenghi & Pradel	2022	PACMPL (OOPSLA)	Empirical taxonomy of real-world software bugs across quantum computing platforms.	Addresses static source-code/compiler bugs rather than dynamic VQC training failures.
[52]	Usandizaga et al.	2026	IEEE ICST	Quantum circuit repair by gate prioritization using fault-localization suspiciousness scores.	Repairs discrete gate mutations against unit tests, not continuous VQC training degradation.
[47]	Chandola et al.	2009	ACM Comput. Surv.	Foundational taxonomy of statistical, information-theoretic, and ML anomaly detection.	General-purpose classical survey; not tailored to quantum gradient/loss telemetry.

Ref	Authors	Year	Venue	Key Contribution	Identified Limitation / Gap Relative to Proposed Study
[48]	Papageorgiou et al.	2022	Front. Manuf. Technol.	Systematic review of machine learning methods for automated root-cause analysis (RCA).	Focused on industrial manufacturing rather than QML optimization landscapes.
[49]	Laptev et al.	2015	ACM SIGKDD	Generic and scalable framework for automated time-series anomaly detection.	Classical temporal monitoring; no coupling to quantum circuit repair arms.
[50]	Gama et al.	2014	ACM Comput. Surv.	Comprehensive survey on concept-drift detection and online model adaptation.	Addresses classical data stream drift rather than quantum hardware/landscape drift.
[41]	Kephart & Chess	2003	IEEE Computer	Establishes Autonomic Computing vision and the Monitor-Analyze-Plan-Execute-Knowledge (MAPE-K) loop.	High-level classical IT architecture; lacks quantum telemetry or bandit repair policies.
[42]	Cheng et al.	2009	Springer LNCS	Software Engineering for Self-Adaptive Systems research roadmap and assurance criteria.	Classical software systems; no digital-twin simulation for quantum circuits.
[43]	Weyns et al.	2012	Springer LNCS	Design patterns for decentralized and feedback-driven self-adaptive control loops.	General software control patterns; lacks probabilistic confidence gating.
[44]	Salehie & Tahvildari	2009	ACM TAAS	Taxonomy of self-healing and self-adaptive software mechanisms and decision criteria.	No application to machine learning training dynamics or quantum circuits.
[53]	Watson	1997	Morgan Kaufmann	Establishes Case-Based Reasoning (CBR) cycle: Retrieve, Reuse, Revise, Retain.	Classical enterprise expert systems; no continuous vector embeddings for QML states.
[54]	Mnih et al.	2015	Nature	Introduces Experience Replay in deep reinforcement learning to break temporal correlations.	General RL control; not structured around failure-diagnosis-repair tuples.
[55]	Johnson et al.	2019	IEEE Trans. Big Data	FAISS: Fast approximate nearest-neighbor similarity search over dense vector embeddings.	Vector indexing infrastructure only; does not define QML failure representations.
[56]	Kritzinger et al.	2018	IFAC-PapersOnLine	Classifies Digital Twin integration levels (Digital Model, Digital Shadow, Digital Twin).	Manufacturing-oriented taxonomy; no quantum noise or VQC simulation.
[57]	Jones et al.	2020	CIRP J. Manuf. Sci.	Systematic characterization of Digital Twins for predictive what-if evaluation.	Fidelity depends on physical model calibration; not applied to QML repair screening.
[58]	Deb et al.	2002	IEEE Trans. Evol. Comput.	NSGA-II: Fast elitist multi-objective optimization for Pareto-front discovery.	Population-based evolutionary search is too expensive for real-time QML repair.
[59]	Roijers et al.	2013	J. Artif. Intell. Res.	Survey of multi-objective sequential decision-making and utility scalarization.	Theoretical framework; not integrated with QML health and overhead metrics.
[60]	Russo et al.	2018	Found. Trends Mach. Learn.	Comprehensive mathematical formulation and regret analysis of Thompson Sampling.	General bandit tutorial; does not incorporate safety filters or digital-twin priors.

Ref	Authors	Year	Venue	Key Contribution	Identified Limitation / Gap Relative to Proposed Study
[62]	Agrawal & Goyal	2013	ICML (PMLR)	Proves O(d sqrt(T ln T)) regret bounds for Linear Contextual Thompson Sampling.	Pure contextual bandit algorithm; lacks diagnostic confidence gating or rollback.
[61]	Li et al.	2010	ACM WWW	Introduces LinUCB contextual bandit framework for personalized action selection.	Deterministic UCB exploration; applied to web recommendation rather than system repair.
[63]	Chapelle & Li	2011	NeurIPS	Empirical demonstration that Thompson Sampling outperforms UCB and greedy heuristics.	Classical ad/news selection; no safety-critical rollback constraints.
[64]	Auer et al.	2002	Machine Learning	Establishes finite-time regret analysis for multi-armed bandit exploration-exploitation.	Context-free bandit formulation; ignores rich failure symptom vectors.
[39]	Li et al.	2017	JMLR	Hyperband: Bandit-based resource allocation for hyperparameter optimization.	Outer-loop pre-training configuration search; cannot heal mid-training failures.
[65]	Lesort et al.	2021	Information Fusion	Comprehensive framework for continual lifelong learning and memory consolidation.	Robotics-focused; not applied to quantum failure-repair knowledge bases.
[66]	Houssein et al.	2022	Expert Syst. Appl.	State-of-the-art survey of quantum machine learning architectures and bottlenecks.	Broad survey; identifies trainability challenges without proposing self-healing.
[67]	Lu et al.	2024	J. Ind. Inf. Integr.	Classifies modern QML challenges across encoding, optimization, and hardware noise.	Treats solutions as isolated techniques rather than a unified runtime loop.

Table 2.1b: Patent Prior-Art Landscape Comparison Matrix

Patent / Pub. No.	Assignee	Year	Main Mechanism	Technical Distinction of Proposed QML Self-Healing Invention
US 10,706,365 B2	D-Wave Systems	2020	Local qubit permutation & weight optimization for quantum annealers.	Hardware topology mapping only; lacks runtime QML failure detection, probabilistic diagnosis, or bandit repair selection.
US 12,067,458 B2	Zapata Computing	2024	Subdividing VQCs into isolated sub-circuits for pre-training initialization.	Static initialization heuristic; lacks confidence-gated diagnosis, failure memory, digital-twin screening, and Contextual Thompson Sampling.
US 12,689,551 B2	Morgan Stanley	2024	Low-latency L1 classical service self-healing failover via ML monitoring.	Classical IT service failover; does not operate on quantum circuits, barren plateaus, or noise-aware simulation proxies.
US 11,343,079 B2	ServiceNow, Inc.	2022	Automated anomaly classification and routing repair in dual-database systems.	Classical database routing; lacks multi-objective quantum circuit evaluation, oscillation filtering, and Bayesian contextual bandits.
US 11,831,729 B2	ServiceNow, Inc.	2023	Predictive ML fault detection over historical IT observability logs.	Pre-failure IT alerting only; lacks closed-loop VQC structural/hyperparameter repair generation and rollback.
WO 2022/081381	UiPath / Int.	2022	AI/ML anomaly detection and self-healing path	Robotic UI automation scripts; completely distinct from hybrid quantum-classical Hilbert-

Patent / Pub. No.	Assignee	Year	Main Mechanism	Technical Distinction of Proposed QML Self-Healing Invention
A1			recalculation for RPA bots.	space optimization landscapes.
IN 202241077370	IARE	2023	Self-healing classical network architecture with AI anomaly detection.	Network packet routing reconfiguration; lacks QML telemetry, entropy confidence gating, and ARSA Thompson Sampling.

2.2 GAPS IDENTIFIED

Synthesizing the state-of-the-art across the 73 surveyed scholarly works and patent prior art reveals five foundational technical gaps (**G1â€“G5**) that prevent reliable autonomous QML training:

Research Gap 1 (G1 â€” Lack of an Integrated Closed-Loop QML Repair Pipeline): Existing QML studies address individual failure modes in complete isolationâ€”e.g., structured initialization for barren plateaus [18, 20], ZNE for hardware noise [25, 26], or natural gradients for ill-conditioned landscapes [12]. No existing system integrates continuous health monitoring, root-cause diagnosis, candidate repair generation, pre-deployment simulation, adaptive selection, and rollback into a unified runtime self-healing loop for VQCs.

Research Gap 2 (G2 â€” Context-Insensitive and Static Repair Selection): Classical training repair systems (such as AutoTrainer [45] and DeepDiagnosis [46]) rely on fixed, deterministic `if-then` mappings from symptom to fix. In QML, however, the effectiveness of a repair (e.g., depth reduction vs. optimizer switch vs. noise mitigation) depends heavily on the joint context vector (qubit count, circuit depth, gradient variance, noise level, and past repair history). Repair selection has not been formulated as a contextual bandit decision problem using Contextual Thompson Sampling.

Research Gap 3 (G3 â€” Uncalibrated Diagnostic Uncertainty & Absence of Confidence Gating): Quantum training symptoms are inherently ambiguous: vanishing gradients can arise from algorithmic over-expressibility [33], global cost functions [19], or hardware decoherence [25]. Existing diagnostic approaches do not quantify posterior diagnostic confidence via entropy gating or trigger multi-pass diagnostic probing before modifying the live circuit.

Research Gap 4 (G4 â€” Absence of Failure Embedding Memory & Continual Experience Reuse): Current QML optimization treats every training failure as a memoryless event. Historical failure-diagnosis-repair outcomes are neither encoded into structured vector embeddings nor retrieved via similarity search to warm-start future repair decisions.

Research Gap 5 (G5 â€” Lack of Safe Pre-Deployment Digital-Twin Screening & Oscillation Prevention): Deploying candidate repairs directly onto an active VQC risks cyclic repair oscillation ($A_1 \rightarrow A_2 \rightarrow A_1$) and severe accuracy collapse. Existing QML frameworks lack an oscillation constraint filter, a noise-aware simulation proxy (Digital Twin) to evaluate multi-objective outcomes prior to deployment, and automated checkpoint rollback.

2.3 OBJECTIVES

To bridge the identified technical gaps (G1â€“G5) and develop a patent-worthy technical invention, the seven measurable objectives (**O1â€“O7**) of this project are defined as follows:

- **Objective O1:** Develop a continuous QML health-monitoring and failure-detection mechanism for VQC training that tracks loss trajectory, accuracy, gradient norms/variance, parameter updates, and circuit noise indicators to compute a composite health score $H_t \in [0,1]$.
- **Objective O2:** Design and implement a probabilistic root-cause diagnosis module with an explicit entropy-based confidence estimator (γ_t) and a Confidence Gate that triggers additional diagnostic passes or lowest-risk fallback when uncertainty is high.
- **Objective O3:** Build a structured Failure Embedding Memory ($\mathcal{K}$) using cosine-similarity k-nearest-neighbor retrieval to reuse historical failure-repair-outcome experiences across training episodes.
- **Objective O4:** Implement constrained candidate repair generation across six repair arms (A_1â€“A_6) coupled with a deterministic Safety and Oscillation Constraint Filter that enforces cooldowns, attempt budgets, and (r_i, r_j, r_i) cycle suppression.
- **Objective O5:** Evaluate surviving candidate repairs prior to live deployment using a Noise-Aware Digital Twin Simulation Proxy and a Multi-Objective Performance Predictor scoring accuracy gain, loss reduction, depth/shot overhead, and stability.
- **Objective O6:** Implement the core **Adaptive Repair Selection Algorithm (ARSA)** based on **Linear Contextual Thompson Sampling (CTS)**, paired with checkpoint-protected deployment, automatic rollback, and online Bayesian posterior updates.
- **Objective O7:** Benchmark the complete self-healing framework against Random, Greedy, and Rule-Based repair selection baselines and conduct 7-component ablation studies across controlled VQC failure scenarios.

Primary Engineering Research Question:

- **Core RQ:** *Can a confidence-aware, context-driven self-healing framework using Contextual Thompson Sampling (ARSA) select suitable repairs for failing VQC training more effectively, safely, and with lower cumulative regret than random, greedy, and rule-based selection policies?*

2.4 PROBLEM STATEMENT

Variational Quantum Circuit (VQC) models trained via hybrid quantum-classical optimization loops suffer from silent, multi-factorial training failuresâ€”including barren plateaus, hardware noise-induced landscape flattening, optimizer instability, excessive circuit depth, and suboptimal feature encodingsâ€”that halt convergence without raising explicit software exceptions. Existing QML mitigation strategies address single failure modes statically prior to training, while classical automated repair tools apply context-blind deterministic rules without accounting for diagnostic uncertainty, quantum noise overhead, or cyclic repair oscillation. Given a VQC model $\mathcal{M}(\boldsymbol{\theta}_t)$ experiencing training degradation $F_t = 1$ at iteration t, the

engineering problem is to design an autonomous closed-loop framework $\mathcal{F}$ that: (1) diagnoses the root-cause probability vector $\mathbf{p}(c|\mathbf{s}_t)$ with calibrated confidence γ_t; (2) retrieves top-k historical cases from Failure Embedding Memory $\mathcal{K}$; (3) filters unsafe or oscillatory candidate repairs $\mathcal{R}'_t \subseteq \{A_1,\dots,A_6\}$; (4) screens candidates via a Noise-Aware Digital Twin $\hat{\mathcal{M}}$; and (5) selects and deploys the optimal repair a^*_t via Contextual Thompson Sampling with checkpoint rollback so as to maximize post-repair recovery rate and minimize cumulative regret $R_T = \sum_{t=1}^T [r^*(\mathbf{x}_t) - r_{a_t}(\mathbf{x}_t)]$.

2.5 PROJECT PLAN

Table 2.2 details the 12-phase execution schedule of the project across the academic timeline, mapping engineering milestones to review deliverables.

Table 2.2: Comprehensive Project Execution Roadmap and Milestones

Phase	Milestone / Technical Activity	Target Timeline	Status / Deliverable
Phase 1	Literature review of 73 papers across 15 themes; prior-art matrices, research gaps & objectives formulation	Week 1	Completed (Review 1)
Phase 2	Define QML failure taxonomy, health telemetry parameters & mathematical formulation	Weeks 1â€“2	Completed (Review 1)
Phase 3	Finalize locked 15-module self-healing architecture, ARSA algorithm & patent claim structure	Weeks 2â€–3	Completed (Review 1)
Phase 4	Implement PennyLane VQC classifier, Continuous Health Monitoring (Module 2) & Failure Detection (Module 3)	Weeks 3â€–5	In Progress (Thin Slice)
Phase 5	Implement Probabilistic Root-Cause Diagnosis (Module 4), Confidence Gate (Module 5) & Failure Memory (Module 6)	Weeks 5â€–6	Immediate Next Step
Phase 6	Implement Candidate Repair Generation (A_1â€“A_6), Safety/Oscillation Filter (Module 8) & Digital Twin Proxy (Module 9)	Weeks 6â€–8	Target for Review 2
Phase 7	Implement Multi-Objective Predictor (Module 10), Contextual Thompson Sampling (Module 11) & Safe Rollback (Module 12)	Weeks 8â€–9	Target for Review 2
Phase 8	Execute baseline comparisons (Random, Greedy, Rule-Based vs. ARSA-CTS) & 7 ablation experiments	Weeks 9â€–10	Target for Review 2
Phase 9	Statistical validation (Wilcoxon signed-rank test), regret curve analysis & overhead benchmarking	Weeks 10â€–11	Post-Review 2
Phase 10	Prepare formal patent specification, Figures 1â€–9 technical drawings & prior-art differentiation package	Weeks 11â€–12	Patent Preparation
Phase 11	System refinement, multi-dataset scaling & comprehensive capstone report compilation	Weeks 12â€–13	Final Report Draft
Phase 12	Final capstone project defense & intellectual property / patent disclosure submission	Weeks 13â€–14	Final Review

CHAPTER 3
TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

- **FR-1 (Continuous VQC Health Telemetry & Anomaly Detection):** The system shall continuously record training loss ℓ_t, classification accuracy a_t, parameter-shift gradient norm $\bar{g}_t$, gradient variance σ^2_g, parameter update velocity $\|\Delta\boldsymbol{\theta}_t\|$, and noise proxy ν_t, computing a normalized composite health score $H_t \in [0,1]$ and raising failure flag $F_t \in \{0,1\}$ when threshold predicates are violated.
- **FR-2 (Probabilistic Diagnosis & Entropy Confidence Gating):** Upon failure detection, the system shall compute a Bayesian posterior distribution $\mathbf{p}(c|\mathbf{s}_t)$ across candidate root causes $c \in \{c_1,\dots,c_K\}$ and evaluate normalized confidence $\gamma_t = 1 - H(\mathbf{p}_t)/\ln K$. If $\gamma_t < \gamma_{\min}$, the Confidence Gate shall trigger up to N_{diag} supplementary diagnostic passes before falling back to the lowest-risk historical repair.
- **FR-3 (Failure Embedding Memory & Case Retrieval):** The system shall project the failure state into a normalized embedding vector $\mathbf{e}_t \in \mathbb{R}^{d_e}$ and retrieve the top-k most similar historical failure-repair records from Knowledge Base $\mathcal{K}$ via cosine similarity.
- **FR-4 (Constrained Repair Generation & Oscillation Filtering):** The system shall instantiate candidate repairs from $\mathcal{A} = \{A_1, A_2, A_3, A_4, A_5, A_6\}$ and prune any candidate that violates maximum attempt counts $N_{\max}(r)$, cooldown windows $\tau_{\text{cool}}(r)$, resource budget B_t, or exhibits cyclic oscillation (r_i, r_j, r_i) within sliding window w_{osc}.
- **FR-5 (Noise-Aware Digital Twin Pre-Deployment Screening):** For each safe candidate $r \in \mathcal{R}'_t$, the system shall clone the current VQC state into a noise-aware simulation proxy $\hat{\mathcal{M}}$, execute a short horizon of τ_{sim} steps, and compute a scalarized multi-objective utility $U(r) \in [0,1]$.
- **FR-6 (Contextual Thompson Sampling & Safe Rollback Deployment):** The system shall sample arm parameters $\tilde{\boldsymbol{\mu}}_a \sim \mathcal{N}(\hat{\boldsymbol{\mu}}_a, v^2\mathbf{B}_a^{-1})$ conditioned on context $\mathbf{x}_t$, deploy $a^*_t = \arg\max_a \mathbf{x}_t^\top \tilde{\boldsymbol{\mu}}_a$ after saving checkpoint $\boldsymbol{\theta}_{\text{backup}}$, automatically roll back if post-repair loss degrades by more than ϵ_{degrad}, and update $\mathbf{B}_{a^*}$, $\hat{\boldsymbol{\mu}}_{a^*}$ and memory $\mathcal{K}$.

3.1.2 Non-Functional Requirements

- **NFR-1 (Bounded Simulation & Diagnostic Latency):** Pre-deployment digital-twin screening shall be bounded to $\tau_{\text{sim}} \le 5$ micro-steps across at most $|\mathcal{R}'_t| \le 4$ filtered repair candidates, keeping self-healing decision overhead below 15% of total VQC training runtime.
- **NFR-2 (Zero-Catastrophic-Regression Safety Guarantee):** Through mandatory pre-repair state checkpointing ($\boldsymbol{\theta}_{\text{backup}}$) and post-repair validation over τ_{eval} steps, any harmful repair intervention shall be deterministically reverted within one evaluation window.
- **NFR-3 (Reproducibility & Deterministic Seeding):** All quantum circuit simulations, injected failure triggers, noise channels, and bandit posterior updates shall log complete random seeds and JSON telemetry traces for 100% experimental reproducibility.
- **NFR-4 (Modular Extensibility):** The 15 modules shall expose standardized Python interfaces so simulator backends (`default.qubit` / `default.mixed`) can be swapped seamlessly with cloud quantum hardware backends (IBM Quantum / Qiskit Runtime).

3.2 FEASIBILITY STUDY

3.2.1 Technical Feasibility

The technical feasibility of the proposed framework is anchored in mature, open-source hybrid quantum-classical libraries. PennyLane [9] provides native automatic differentiation and parameter-shift gradient access for VQC monitoring, while `default.mixed` and Qiskit Aer noise models enable accurate depolarizing and amplitude-damping channels for the Noise-Aware Digital Twin (Module 9). Furthermore, Linear Contextual Thompson Sampling [62] requires only $d \times d$ matrix inversions ($\mathbf{B}_a^{-1}$ with $d \approx 24$), which execute in under 2 milliseconds on a standard CPU using rank-1 Sherman-Morrison updates.

3.2.2 Economic Feasibility

The project incurs zero capital expenditure or proprietary licensing cost. All VQC training runs, controlled failure injections, digital-twin simulations, and bandit policy evaluations execute on standard local workstation hardware or free-tier Google Colab runtimes using open-source software (Python, PennyLane, NumPy, SciPy, PyTorch, and FAISS), eliminating expensive commercial quantum hardware queue costs during prototyping and validation.

3.2.3 Social & Ethical Feasibility

All experiments utilize publicly available, non-sensitive benchmark datasets (Iris, Wine, Breast Cancer). By preventing silent convergence failures, redundant quantum circuit re-executions, and wasted computational energy, the self-healing framework promotes sustainable, reproducible, and trustworthy quantum machine learning engineering.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

- **Host Processor:** Multi-Core x86_64 Processor (Intel Core i7 / Xeon @ 2.40+ GHz, 8+ Threads for parallel candidate repair simulation)
- **Compute Accelerator:** NVIDIA CUDA-enabled GPU (Tesla T4 / RTX 16 GB VRAM) or High-Speed CPU Statevector/Density-Matrix Simulator
- **System Memory (RAM):** 16 GB DDR4/DDR5 High-Speed System RAM
- **Storage:** 50 GB NVMe SSD Storage for checkpoint snapshots, telemetry logs, and Failure Embedding Memory ($\mathcal{K}$)

3.3.2 Software Specification

- **Operating System:** Windows 11 (64-bit) / Ubuntu 22.04 LTS
- **Programming Language:** Python 3.10+
- **Quantum ML & Simulation Frameworks:** PennyLane 0.35+ (`default.qubit`, `default.mixed`), PennyLane-Qiskit Noise Plugin
- **Numerical & Bandit Optimization Libraries:** NumPy 1.26+, SciPy 1.12+, PyTorch 2.2+
- **Vector Similarity & Evaluation Tooling:** Scikit-learn 1.4+, FAISS-CPU (for Failure Embedding Memory retrieval), Matplotlib/Seaborn

CHAPTER 4

DESIGN APPROACH AND DETAILS

4.1 SYSTEM ARCHITECTURE

Figure 4.1 delineates the locked 15-module closed-loop system architecture of the **Adaptive Self-Healing Framework for Quantum Machine Learning**. Unlike conventional static QML pipelines, the architecture wraps the active VQC training loop inside a continuous autonomic control cycle consisting of five functional tiers: (1) Continuous Health Monitoring & Failure Detection (Modules 1â€“3), (2) Probabilistic Root-Cause Diagnosis & Confidence Gating (Modules 4â€“5), (3) Failure Embedding Memory & Constrained Candidate Repair Generation (Modules 6â€“8), (4) Noise-Aware Digital Twin Simulation & Multi-Objective Prediction (Modules 9â€“10), and (5) Contextual Thompson Sampling Selection (ARSA), Safe Checkpoint Deployment & Continual Online Learning (Modules 11â€“15).

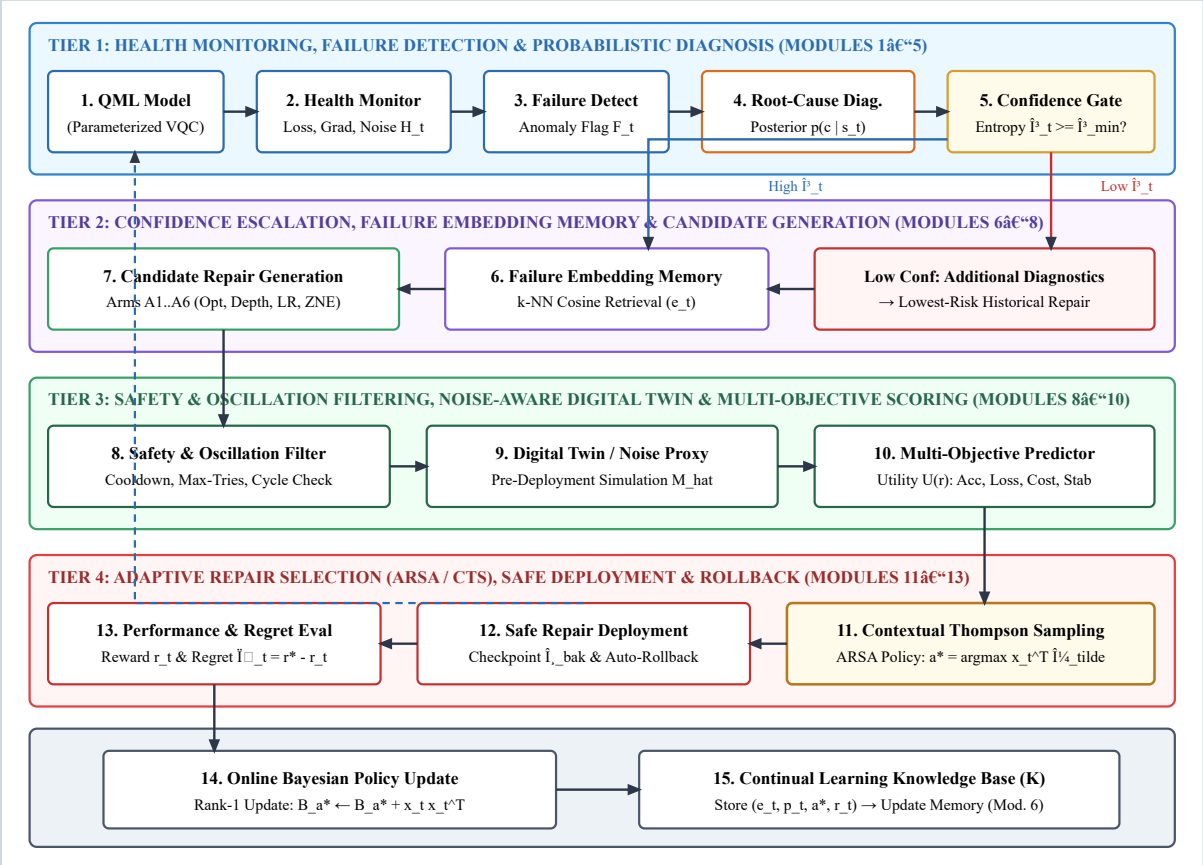


Figure 4.1: Proposed 15-Module Closed-Loop System Architecture

The architectural design departs fundamentally from open-loop QML training by introducing two feedback loops: (1) the **Pre-Deployment Verification Loop** (Modules 5 → 8 → 9 → 10), which screens candidate repairs against oscillation rules and evaluates them on a noise-aware digital twin prior to touching the live circuit; and (2) the **Continual Bandit Learning Loop** (Modules 11 → 12

→ 13 → 14 → 15), which updates both the Contextual Thompson Sampling posterior parameters $(\mathbf{B}_a, \hat{\boldsymbol{\mu}}_a)$ and the Failure Embedding Memory $\mathcal{K}$ after every repair episode.

4.2 DESIGN AND FLOW

4.2.1 Process Flow & Decision Diagram

Figure 4.2 presents the complete operational execution logic and decision flow of the self-healing pipeline, highlighting the three sequential safety gates: (i) the **Diagnosis Confidence Gate** ($\gamma_t \ge \gamma_{\min}$), (ii) the **Safety & Oscillation Constraint Gate** ($\text{Safe}(r,t)=1$), and (iii) the **Post-Deployment Rollback Gate** ($\ell_{t+\tau} \le \ell_t + \epsilon_{\text{degrad}}$).

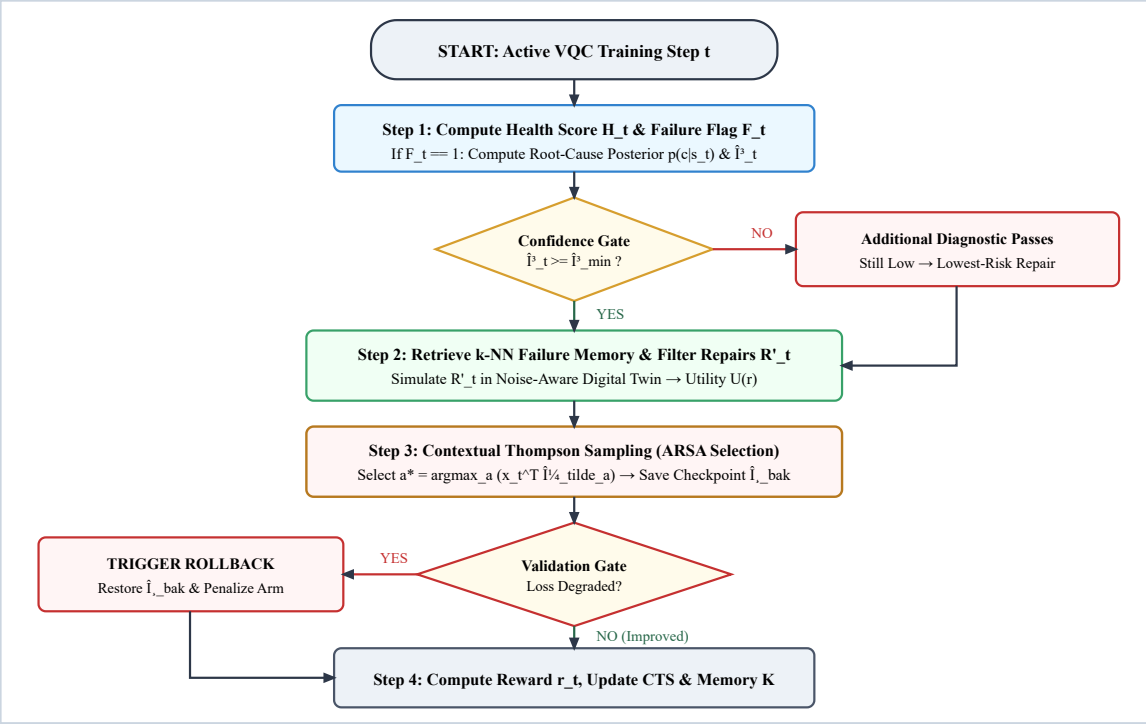


Figure 4.2: Execution Process & Safety Gate Decision Flowchart

4.2.2 Patent Concept (ARSA) & Supervisor Approval Status

The core candidate invention protected by this project is the **Adaptive Repair Selection Algorithm (ARSA)**. Rather than claiming generic "AI for quantum circuits," ARSA defines a specific, technically concrete closed-loop machine control process wherein: (1) entropy-gated probabilistic fault diagnosis governs diagnostic depth; (2) historical failure embeddings warm-start candidate generation; (3) an oscillation filter and noise-aware digital twin screen candidate repairs prior to hardware/circuit modification; and (4) Contextual Thompson Sampling selects the repair action conditioned on the real-time quantum-classical telemetry vector $\mathbf{x}_t$. The project title, Review-1 presentation, and technical architecture have been formally reviewed and approved on

VTOP by project supervisor **Dr. Nithya NS** (Associate Professor Grade 2, SCOPE, VIT; Project ID: **20127IPG02**, Reg. No: **22MID0324**, Date: **29-09-2026**).

CHAPTER 5
METHODOLOGY AND TESTING

5.1 MODULE DESCRIPTION

5.1.1 Modules 1â€“3: VQC Model, Continuous Health Monitoring & Failure Detection

Module 1 executes an n-qubit parameterized quantum circuit $|\psi(\mathbf{x},\boldsymbol{\theta})\rangle = U_L(\boldsymbol{\theta}_L)\cdots U_1(\boldsymbol{\theta}_1)S(\mathbf{x})|0\rangle^{\otimes n}$. At each training iteration t, Module 2 computes normalized health indicators over sliding window w: gradient health $h_1(t) = \min(1, \bar{g}_t/\epsilon_g)$, loss progress $h_2(t) = \sigma(-\Delta\ell_t/\tau_\ell)$, loss stability $h_3(t) = \exp(-\text{Var}(\ell_{t-w:t})/\tau_s)$, and accuracy progress $h_4(t)$. The composite health score and failure trigger are defined as:

$$H_t = \Sigma_{i=1..4}\, \alpha_i \cdot h_i(t) \in [0, 1], \qquad F_t = I[H_t < H_{min}] \vee I[\bar{g}_t < \varepsilon_g \text{ for } w_g \text{ steps}] \vee I[|\Delta\ell_t| < \varepsilon_\ell \text{ for } w_\ell \text{ steps}]$$

5.1.2 Modules 4â€“5: Probabilistic Root-Cause Diagnosis & Confidence Gate

When $F_t = 1$, Module 4 assembles the symptom vector $\mathbf{s}_t = [\bar{g}_t, \sigma^2_g, \ell_t, \Delta\ell_t, a_t, L, n, \eta, \nu_t]^\top$ and computes the posterior probability over K root causes (Barren Plateau c_1, Excessive Learning Rate c_2, Noise Degradation c_3, Poor Feature Map c_4, Excessive Depth c_5, Bad Initialization c_6) along with normalized Shannon entropy confidence γ_t:

$$p(c_j \mid s_t) = [\, p(s_t \mid c_j) \cdot p(c_j) \,] \,/\, [\, \Sigma_{k=1..K}\, p(s_t \mid c_k) \cdot p(c_k) \,], \qquad \gamma_t = 1 + [\, \Sigma_{j=1..K}\, p(c_j \mid s_t) \ln p(c_j \mid s_t) \,] \,/\, \ln(K) \in [0, 1]$$

If $\gamma_t < \gamma_{\min}$, Module 5 (Confidence Gate) executes additional diagnostic probes (extended shot sampling and Hessian/gradient sensitivity probing). If γ_t remains below $\gamma_{\min}$, the gate selects the lowest-risk historical repair from memory.

5.1.3 Modules 6â€“8: Failure Embedding Memory, Candidate Generation & Safety Filter

Module 6 encodes the failure context into normalized vector $\mathbf{e}_t = \text{normalize}([\mathbf{s}_t; \mathbf{p}_t; \gamma_t; \boldsymbol{\mu}_t])$ and retrieves the top-k historical cases from $\mathcal{K}$ using cosine similarity $\text{sim}(\mathbf{e}_t, \mathbf{e}_i) = (\mathbf{e}_t^\top \mathbf{e}_i)/(\|\mathbf{e}_t\|\|\mathbf{e}_i\|)$. Module 7 activates candidate repairs from $\mathcal{A} = \{A_1, \dots, A_6\}$, and Module 8 filters out any candidate violating the safety and oscillation predicate:

$$R'_t = \{\, r \in R_t \mid n_{attempts}(r, t) < N_{max}(r) \wedge (t - t_{last}(r)) > \tau_{cool}(r) \wedge RepairSeq(t-2 : t) \neq (r', r, r') \,\}$$

5.1.4 Modules 9â€“10: Noise-Aware Digital Twin Simulation & Multi-Objective Prediction

For every safe candidate $r \in \mathcal{R}'_t$, Module 9 instantiates a virtual digital twin $\hat{\mathcal{M}} = (\mathcal{C}, \mathcal{N}, \mathcal{O}, \boldsymbol{\theta}_t)$ incorporating the current circuit topology and depolarizing/readout noise model $\mathcal{N}$, simulating τ_{sim} steps to estimate post-repair metrics. Module 10 computes the scalarized multi-objective utility $U(r)$ across accuracy improvement u_1, loss reduction u_2, negative circuit/shot cost u_3, and gradient stability u_4:

$$\mathbf{U(r) = w_1 \cdot \hat{u}_1(r) + w_2 \cdot \hat{u}_2(r) + w_3 \cdot \hat{u}_3(r) + w_4 \cdot \hat{u}_4(r) \in [0, 1]}$$

5.1.5 Modules 11â€“15: Contextual Thompson Sampling (ARSA), Safe Deployment & Algorithms

Module 11 constructs the d-dimensional context vector $\mathbf{x}_t = [\mathbf{e}_t; \mathbf{p}_t; \gamma_t; U(\mathcal{R}'_t)] \in \mathbb{R}^d$ and models each repair arm's reward as $r_t = \mathbf{x}_t^\top \boldsymbol{\mu}_a + \epsilon_t$. Maintaining precision matrix $\mathbf{B}_a = \mathbf{X}_a^\top \mathbf{X}_a + \lambda\mathbf{I}$ and mean estimator $\hat{\boldsymbol{\mu}}_a = \mathbf{B}_a^{-1}\mathbf{f}_a$, the complete execution is formalized in Algorithms 1 and 2 below.

Algorithm 1: Confidence-Gated Diagnosis, Memory Retrieval & Safety-Constrained Digital Twin Screening

---

Input: Symptom vector $\mathbf{s}_t$, VQC state θ_t, Failure Memory K, Candidate Arms $A = \{A_1..A_6\}$, Threshold $\gamma_{\min}$

Output: Context vector $\mathbf{x}_t$, Filtered & Simulated Safe Repair Set R'_t, Utility Scores $\{U(r)\}$

Compute posterior $\mathbf{p}_t(c \mid \mathbf{s}_t)$ via Bayes' rule and entropy confidence $\gamma_t = 1 - H(\mathbf{p}_t) / \ln(K)$

if $\gamma_t < \gamma_{\min}$ **then**

  Execute supplementary diagnostic pass (increase shot budget & parameter-shift sensitivity probe)

  Recompute $\mathbf{p}_t$ and γ_t; **if** still $\gamma_t < \gamma_{\min}$ **then** flag lowest-risk historical repair

end if

Encode failure embedding $\mathbf{e}_t \leftarrow$ Normalize($[\mathbf{s}_t; \mathbf{p}_t; \gamma_t; \mu_t]$) and retrieve top-$k$ nearest cases from K

Filter safe candidates: $R'_t \leftarrow \{ r \in A \mid \text{Attempts}(r) < N_{\max}$ AND CooldownExpired(r) AND NOT Oscillating(r) $\}$

for each candidate repair $r \in R'_t$ **do**

  Simulate τ_{sim} steps on Noise-Aware Digital Twin $\hat{M}(\theta_t, r)$ and compute multi-objective score $U(r)$

end for

return Context $\mathbf{x}_t = [\mathbf{e}_t; \mathbf{p}_t; \gamma_t; U(R'_t)]$, Safe Set R'_t

---

Algorithm 2: Adaptive Repair Selection Algorithm (ARSA) via Contextual Thompson Sampling & Safe Rollback

---

Input: Context $\mathbf{x}_t \in \mathbb{R}^d$, Safe Repairs R'_t, Arm Posteriors $\{(\mathbf{B}_a, \mathbf{f}_a, \hat{\boldsymbol{\mu}}_a)\}_{a \in A}$, Exploration scale v, Rollback tolerance $\varepsilon_{\text{degrad}}$

Output: Healed VQC state $\theta_{t+\tau}$, Updated Policy Posteriors, Updated Knowledge Base K

for each safe repair arm $a \in R'_t$ **do**

  Sample parameter vector $\tilde{\boldsymbol{\mu}}_a \sim \mathcal{N}(\hat{\boldsymbol{\mu}}_a, v^2 \mathbf{B}_a^{-1})$

  Compute sampled contextual expected reward: $\tilde{r}_a = \mathbf{x}_t^\top \tilde{\boldsymbol{\mu}}_a + \eta_{DT} \cdot U(a)$

end for

Select optimal repair arm: $a^* \leftarrow \text{argmax}_{a \in R'_t} \tilde{r}_a$

6: Save checkpoint (θ_{backup}, $OptState_{backup}$) and deploy repair a* to live VQC model *M*
7: Train for τ_{eval} validation steps and observe post-repair loss $\ell_{t+\tau}$, accuracy $a_{t+\tau}$, and health $H_{t+\tau}$
8: **if** $\ell_{t+\tau} > \ell_t + \varepsilon_{degrad}$ **then**
9: Restore (θ_{backup}, $OptState_{backup}$) // Automatic Rollback Protection
10: Assign penalty reward $r_t \leftarrow r_{min}$ and place arm a* on cooldown
11: **else**
12: Compute realized reward $r_t = \alpha_r \Delta a_t - \beta_r \Delta \ell_t + \gamma_r \Delta H_t - \delta_r$ Overhead(a*)
13: **end if**
14: Update posterior: $\mathbf{B}_{a^*} \leftarrow \mathbf{B}_{a^*} + \mathbf{x}_t \mathbf{x}_t^\top$, $\mathbf{f}_{a^*} \leftarrow \mathbf{f}_{a^*} + r_t \mathbf{x}_t$, $\hat{\boldsymbol{\mu}}_{a^*} \leftarrow \mathbf{B}_{a^*}^{-1} \mathbf{f}_{a^*}$
15: Append tuple ($\mathbf{e}_t$, $\mathbf{p}_t$, γ_t, a*, r_t) to Continual Learning Knowledge Base *K*

5.2 EXPERIMENTAL SETUP AND EVALUATION PLAN

5.2.1 Datasets, Data Strategy & Failure Injection Scenarios

Table 5.1 summarizes the benchmark datasets, VQC configurations, and reproducible failure-injection scenarios designed to validate the four data streams of the framework (QML Benchmark Data, Training Health Telemetry, Injected/Observed Failure Data, and Post-Repair Outcome Data).

Table 5.1: Experimental Datasets, VQC Topologies, and Controlled Failure Injection Scenarios

Scenario ID	Dataset & Qubits	Injected Failure Condition	Observable Telemetry Signature	Primary Target Repair Arm(s)
Scenario S1 (Primary Slice)	Iris (4 Qubits, Angle Encoding)	Excessive Learning Rate ($\eta = 0.50$)	High loss variance σ^2_ℓ, oscillating $\Delta\ell_t$ sign flips	A_4 (LR Reduction) & A_1 (Switch to SPSA/Adam)
Scenario S2 (Barren Plateau)	Iris / Wine (4â€“8 Qubits, 20 Layers)	Excessive Circuit Depth ($L = 20$ StronglyEntangling)	Exponential gradient collapse $\bar{g}_t < 10^{-5}$, flat loss	A_2 (Circuit-Depth Reduction) & A_6 (Identity Init)
Scenario S3 (NISQ Noise)	Wine (4 Qubits, Depolarizing)	Depolarizing Gate Noise ($p_{\text{depol}} = 0.05$)	Noise-induced gradient suppression & fidelity drop	A_5 (ZNE Noise Mitigation) & A_2 (Depth Pruning)
Scenario S4 (Encoding Fault)	Breast Cancer PCA (4 Qubits)	Suboptimal / Constant Feature Scaling	Stagnant training accuracy $a_t \approx 0.50$ despite non-zero $\bar{g}_t$	A_3 (Feature-Map Re-Uploading Modification)
Scenario S5 (Trap / Init)	Iris (4 Qubits, Zero/Bad Init)	Adversarial Local Minimum / Collinear Initialization	Immediate zero-gradient plateau at high loss ℓ_0	A_6 (Structured Warm Restart / Reinitialization)

5.2.2 Baselines, Ablation Studies & Evaluation Metrics

The proposed **ARSA (Contextual Thompson Sampling)** policy will be rigorously benchmarked against three primary baselinesâ€”(1) **Random Repair Selection**, (2) **Context-Free Greedy Repair Selection**, and (3) **Static Rule-Based Selection** (AutoTrainer-style deterministic mapping)â€”plus seven systematic ablation variants (removing Failure Memory, Confidence Gate, Digital Twin, Safety/Oscillation Filter, Context Vector, Bandit Exploration, and Online Posterior Updates). Performance will be quantified across five metric categories:

- **Failure Detection & Diagnosis Precision:** Detection F1-score, detection latency (steps from injection to trigger), root-cause top-1 accuracy, and Expected Calibration Error (ECE) of confidence γ_t.
- **Self-Healing Recovery Rate & Final Accuracy:** Percentage of failing VQC runs successfully restored to convergence threshold and post-repair test classification accuracy.
- **Cumulative Regret & Reward Trajectory:** Empirical cumulative regret $R_T = \sum_{t=1}^T (r^*_t - r_t)$ over sequential failure episodes demonstrating continual policy improvement.
- **Safety & Stability Metrics:** Rollback invocation rate (%), oscillation incidence rate (%), and average repair attempts per failure episode.
- **Computational Overhead:** Wall-clock simulation latency and quantum circuit execution overhead relative to standard unmonitored training.

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